

Lecture 8: Real-Time Rendering Technology

“Why do some VR scenes look like real photos, while others look like cartoons?”

Show (or describe):

- A photorealistic VR environment
- A low-quality game environment

Explain:

The difference comes from **rendering techniques + GPU power + asset quality (like photogrammetry models)**.



5–20 min → Rasterization vs Ray Tracing



Rasterization (Traditional real-time method)

Explain simply:

Rasterization converts 3D triangles into 2D pixels quickly.

Steps:

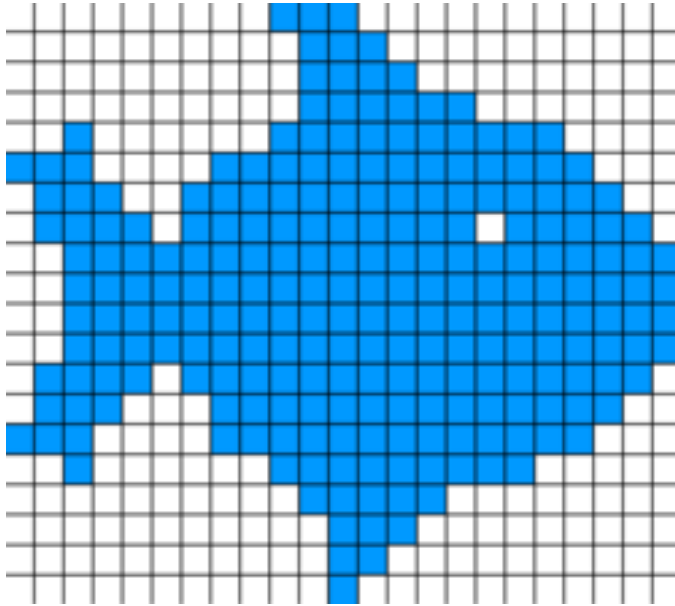
- Objects broken into triangles
- Vertices projected onto screen
- Pixels filled with color using shaders

Advantages:

- Very fast
- Used in games, mobile AR, VR headsets
- Good for real-time performance

Limitations:

- Fake lighting
- Reflections approximated
- Shadows are less realistic



✓ Ray Tracing (Physically realistic method)

Explain simply:

Ray tracing simulates how real light behaves.

How it works:

- Shoots rays from camera
- Rays bounce off surfaces
- Calculates reflections, refractions, shadows

Advantages:

- Ultra realistic
- Cinematic quality
- Better for photogrammetry realism

Limitations:

- Computationally expensive
- Needs powerful GPUs (RTX, AI cores)



Simple Comparison Table (use on board/slide)

Rasterization	Ray Tracing
Faster	Slower
Approximate lighting	Physically accurate lighting
Used in mobile VR	Used in high-end VR
Game-like realism	Near real-world realism

GPU Architecture Overview

Explain why GPU exists:

CPUs are good at logic, GPUs are good at massive parallel math.

Basic GPU components:

- Thousands of small cores
- Streaming Multiprocessors (SMs)
- VRAM (Video Memory)
- Shader Cores

- **Ray Tracing Cores (modern GPUs)**

Analogy:

CPU = 1 brilliant professor

GPU = 10,000 fast students working together

Explain relevance to AR/VR:

- Each frame contains millions of pixels
- Each pixel can be computed in parallel
- Headsets require 90–120 FPS → only GPUs can handle this

Connecting Photogrammetry with Rendering

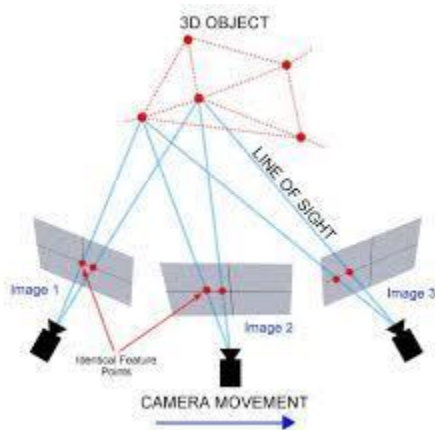
This is where your lecture becomes advanced and impressive.

What is Photogrammetry?

Photogrammetry = creating 3D models from real-world photos.

Pipeline:

1. Take many photos of object
2. Software reconstructs 3D mesh
3. Texture maps generated
4. Imported into Unity/Unreal/VR environment



Why Rendering Matters for Photogrammetry

Photogrammetry produces:

- Very high polygon meshes
- Ultra high-resolution textures
- Realistic surface details

Now connect:

Photogrammetry Asset	Rendering Requirement
Millions of polygons	GPU must handle heavy geometry
Realistic textures	Needs advanced shaders
Natural lighting	Ray tracing gives best realism
Real-time VR usage	Rasterization + optimization needed

Explain:

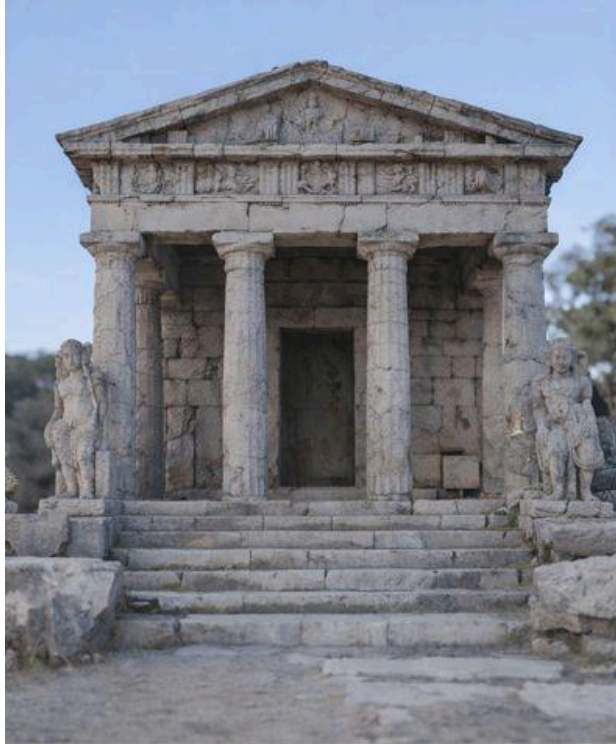
Photogrammetry gives realism in data, but **rendering technology determines whether it feels real inside VR.**

Example explanation to tell students:

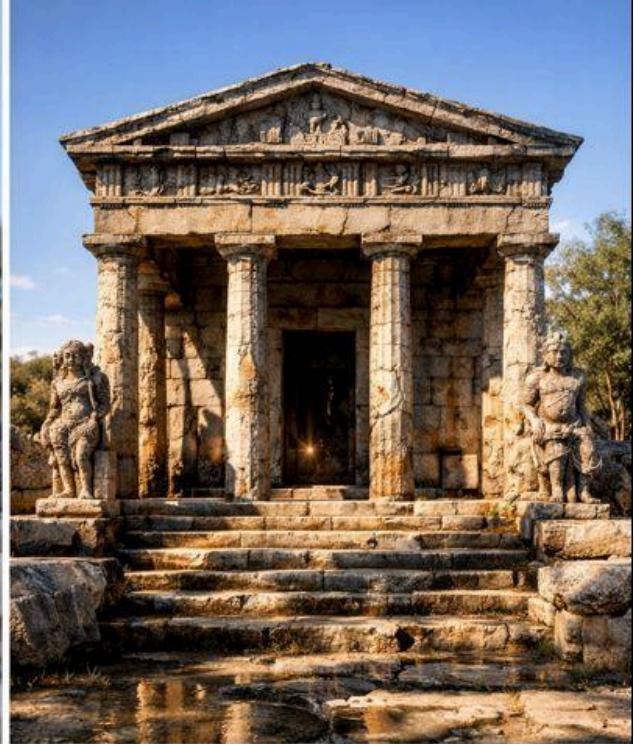
A photogrammetry-scanned temple looks real.

But without good lighting, shadows, reflections, and GPU power — it still feels fake.

Ray tracing + good GPU = true realism.



Basic Rasterization



Ray Tracing for Realism

50–55 min → Real-World Use Case Discussion

Discuss examples:

- Google Earth VR → photogrammetry + GPU rendering
- Medical VR → scanned organs + real-time rendering
- Cultural heritage VR → scanned monuments

Recap + Quick Questions

Recap:

- Rasterization = fast, real-time friendly
- Ray tracing = realistic, GPU expensive
- GPU architecture enables parallel rendering
- Photogrammetry creates realistic models, but rendering determines experience

Quick oral questions:

- Why is ray tracing slower than rasterization?
- Why do photogrammetry models require optimization?
- Why is GPU more critical than CPU in VR?